

Exp. and Log. fns.

✓ The real exp. fn
 $\exp: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto e^x = \exp(x)$
can be characterized in several ways.

① As a power series.

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Later!!

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad **$$

✓ We henceforth get it that

$$e^0 = \exp(0) = 1$$

$$\text{and } e^1 = \sum_{k=0}^{\infty} \frac{1}{k!}$$

✓ Recall:

$$\frac{d}{dx} (f \pm g) = \frac{df}{dx} \pm \frac{dg}{dx}$$

✓ Also, a term by term differentiation of e^x reveals that

$$\boxed{\frac{d(e^x)}{dx} = e^x \quad \forall x \in \mathbb{R}}$$

How and why?

$$\frac{d(x^n)}{dx} = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = \frac{d}{dx} \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3 \cdot 2!} + \frac{x^4}{4 \cdot 3!} + \dots \right)$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = e^x.$$

② As a limiting value (or as a limit).

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$

Later!!

(on indeterminate forms).

1^∞

③ Via the natural log. fn.

Recall

Properties of exponents/indices.

eg

$$* e^x \cdot e^y = e^{x+y}$$

$$* e^{-x} = \frac{1}{e^x}$$

$$* (e^x)^y = e^{xy}$$

$$* \frac{e^x}{e^y} = e^{x-y}$$

(forget about proofs!)

The (natural) log fn

✓ Recall

$$\boxed{y = \log_a x \iff x = a^y} ***$$

The following properties hold (due to ***)

$$1) \log_a pq = \log_a p + \log_a q \quad *$$

$$2) \log_a \left(\frac{p}{q}\right) = \log_a p - \log_a q$$

$$3) \log_a p^r = r \log_a p, \text{ etc}$$

where p, q are two real numbers and r is any real number.

Proof (1)

We want to show that $\log_a pq = \log_a p + \log_a q$.

Let $x = \log_a p$.

$$\Updownarrow$$

$$p = a^x$$

and

let $y = \log_a q$

$$\Updownarrow$$

$$q = a^y$$

Then

$$pq = a^x \cdot a^y \implies pq = a^{x+y}$$

$$\log_a pq \stackrel{\Updownarrow}{=} \textcircled{x} + y \quad \square$$

Recall:

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

Later!

✓ Also, recall that

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

← power rule of integration.

What if $n = -1$? or what is

we don't know :)

$$\int x^{-1} dx?$$

We just hope that it turns out to be $\ln x$..

Definition

For all $x > 0$, we define

$$\int_1^x \frac{1}{t} dt = \ln x$$

✓ The expression $\ln x$ is called the natural logarithm of x .

✓ x has to be positive since $\int_1^x \frac{1}{t} dt$ does not exist when $x \leq 0$.

✓ The FTC implies that

$$\frac{d}{dx} \left(\int_1^x \frac{1}{t} dt \right) = \frac{d}{dx} (\ln x)$$

$$\frac{1}{x} = \frac{d}{dx} (\ln x)$$

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

Let $a = e$.

Then

$$\frac{d}{dx} (\log_e x) = \frac{1}{x \ln e}$$

✓ We can deduce that

$$\ln 1 = \int_1^1 \frac{1}{t} dt$$

$$\boxed{\ln 1 = 0}$$

$$\boxed{\ln x = \log_e x}$$

$$\log_a p^r = r \log_a p$$

✓ And

$$\ln e^x = \int_1^{e^x} \frac{1}{t} dt$$

$$x \ln e = \int_1^{e^x} \frac{1}{t} dt$$

$$x = \int_1^{e^x} \frac{1}{t} dt$$

$$y = \log_a a \quad (=1)$$

$$a^1 = a^y \iff y=1$$

✓ We arrive at deducing that

$$\boxed{\ln e = 1}$$

Theorem

To every real number x , there's a unique (corresponding) positive real number y s.t

$$\ln y = x \iff y = e^x.$$

$$\log_a y = x$$
$$\Downarrow$$
$$y = a^x$$